

Valentin Legras

Dear Hiring Team,

About myself

I am a Senior Data Scientist with 7+ years of pharmaceutical data science experience across clinical pharmacometrics, PK/PD workflows, and reusable scientific software. I build reproducible R and Python analysis tools, machine learning models, and data products that help scientists turn complex clinical and experimental data into decisions. I am applying for the Principal Machine Learning Scientist, Frontier Research role because it sits at the intersection of rigorous machine learning, open science, and drug-development impact.

Why I fit the role

The role calls for scientists who can frame difficult biological questions, design experiments, write reusable code, evaluate results, communicate clearly, and collaborate across a distributed research organization. My strongest fit is on the applied scientific software and drug-development side:

- **Drug-development data science:** I have delivered PK/PD, pharmacometric, and clinical data workflows across Oncology and Neuroscience Phase 1 to 3 studies, partnering with clinical, statistical, and pharmacometric teams.
- **Reusable scientific code:** I founded and led aNCA, an open-source R package now used by 120+ clinical and pre-clinical scientists, establishing version control, testing, documentation, code review, CI/CD, and regulatory-grade development standards.
- **Machine learning and evaluation:** I have built and evaluated machine learning models including Logistic Regression, Gradient Boosting, and Random Forest, and used explainable-AI techniques such as feature importance and SHAP to make model behavior interpretable.
- **Experiment-to-impact workflow:** I design reproducible pipelines from data acquisition and preparation through analysis, review, reporting, and stakeholder decision making, reducing manual effort by 90 percent in core workflows.
- **Scientific communication:** I present technical work externally at PHUSE and PAGE, translate scientific requirements into engineering plans, and lead cross-functional teams around shared analytical tools.

What I would bring

My background is strongest in applied pharmaceutical data science, scientific software engineering, and practical machine learning rather than in theoretical generative modeling research. That said, I bring a rare combination of domain fluency, hands-on implementation, open-source product ownership, and the ability to scale reproducible analytical workflows in a regulated scientific environment. I would contribute immediately to experiment design, reusable code, evaluation discipline, collaboration with scientists, and translation of machine learning methods into tools that support medicine development.

I would welcome the chance to discuss where this applied ML and scientific software profile could best support your Frontier Research agenda.

Sincerely

Valentin Legras

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